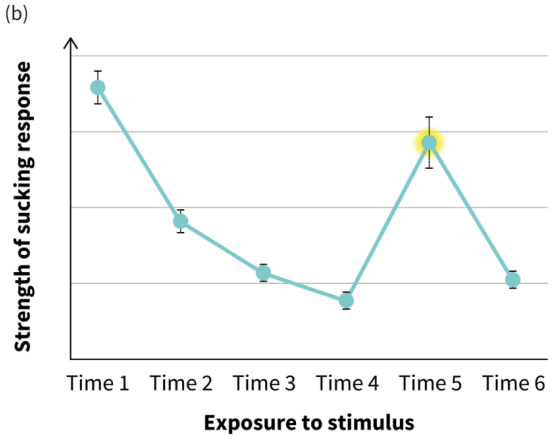
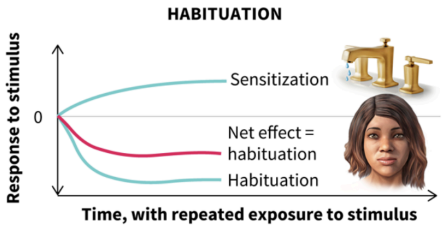
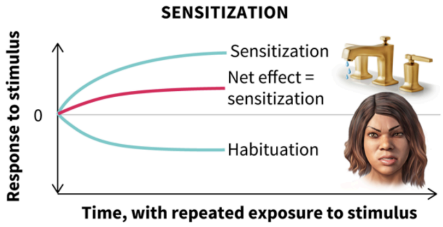
6.1 — Basic Forms of Learning: Associative and Nonassociative

- **learning:** process by which an organism's experience produces enduring changes in both the brain and behavior (*e.g.* improved skill / understanding)
    - adapting environment, adapt to environment
- **associative learning:** make *connections* (*e.g.* dark clouds -> rain, umbrella -> dry)
- **nonassociative learning:** change in how much / little we respond to a single stimulus with experience
    - **habituation:** reflexive response to repeated stimulus becomes weaker
        - waste of energy to attend to unchanging stimuli
        - **dishabituation:** recovery of initial startle (*e.g.* next cycle of car alarm w/ some novel differences) in learned habituation (as opposed to sensory adaptation)



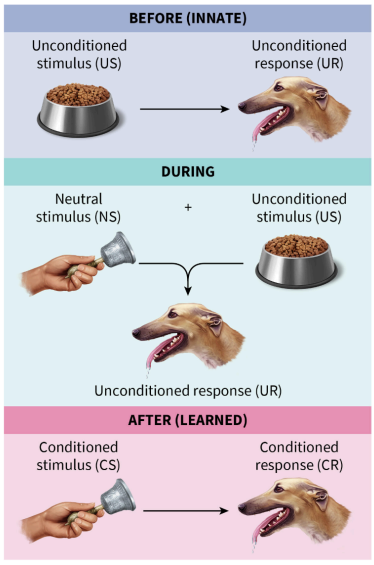
    - **sensitization:** becomes stronger
        - when repeated stimuli means more attention is needed (noxious stimuli)
    - method: study eye movement / looking time / where they look
    - dual-process theory of nonassociative learning: both habituation & sensitization always at work, distinct opposing processes
        - which one wins depends on state of psychological arousal
        - really alert (*e.g.* danger): sensitization > habituation





6.2 — Nuts and Bolts of Classical Conditioning

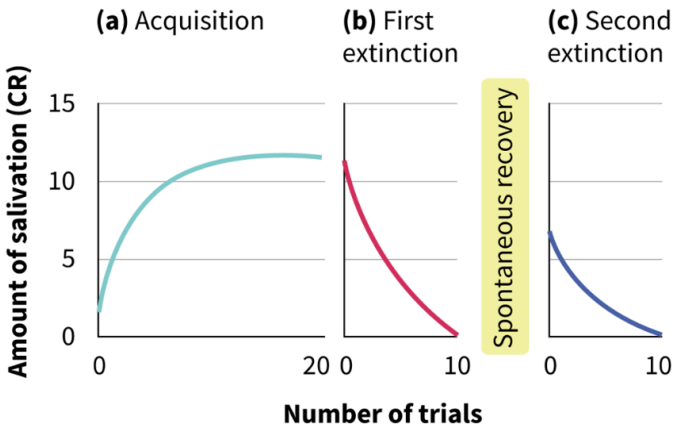
- **associative learning:** connection made between 2 events
- *conditioning*: process that enables us to learn associations

    - changes in behavior that are conditional (dependent) on associations made
- **classical conditioning:** passive form of associative learning, involuntary response -> new stimulus
    - Ivan Pavlov: dogs associating sights / sounds with food
    - *conditioned reflex* through classical conditioning
    - **unconditioned stimulus (US):** produces a response without prior learning
    - **unconditioned response (UR):** response that is automatically generated by US
    - Pavlov: food (US) produced saliva (UR)
    - **conditioned stimulus (CS):** A stimulus that has no prior positive or negative association but comes to elicit a response after being associated with the unconditioned stimulus.
    - **conditioned response (CR):** A response that occurs in the presence of the conditioned stimulus after an association between the conditioned and unconditioned stimulus is learned.
    - Pavlov: UR & CR = same response once the association between CS (Pavlov) and US (food) was made
    - 

6.3 — What Classical Conditioning Teaches Us About Learning

- fundamental processes
- **acquisition:** initial learning of US-CS link (pairing of food & bell)
    - learning increases rapidly as CS evokes CR more reliably -> association fully learned
- **generalization:** tendency to respond to stimuli similar to CS
    - greater similarity, more likely to be generalized
    - sometimes: *overgeneralization*
- **discrimination:** learn to respond to particular stimulus but not others, preventing overgeneralizations
- **extinction:** CR is weakened in response to CS if infrequently presented in absence of US
    - ring bell (CS) but no longer brought food (US) -> eventually stop salivating
    - doesn't imply complete unlearning
- **spontaneous recovery:** extinct behavior reappears after delay
    - many subsequent delay periods -> extinction prevail & spontaneous recovery would no longer occur

    - association not fully forgotten, just suppressed

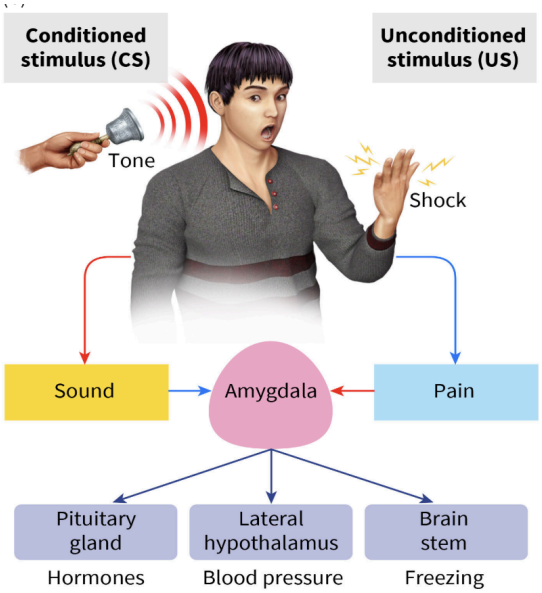


    - cravings of cigarette smokers
    - suggests classical conditioning is somewhat permanent
- relearning association after extinction is faster than original conditioning
- rapid relearning: *savings* (learning stored away in brain bank)

6.4 — Classical Conditioning Provides the Power of Prediction
- success of classical conditioning depends on closeness in time (*contiguity*) of CS and US
- *contingency* (predictiveness) is more important than contiguity
- learning: predictive future based on past
    - co-occur, not worthwhile because limited ability for prediction
- dopamine-secreting neurons
    - dopamine first secreted to US
    - then after association, only after CS
    - neurons responding to PREDICTION of pleasure
    - US not followed by CS, then *prediction error* occurs -> signal brain to revise prediction -> basis for extinction
- conditioning from familiar stimuli harder than unfamiliar
- **blocking:** tone paired with mild shock (US), then light added to the tone (presented simultaneously), then:
    - tone elicits CR while light does not because tone first then light doesn't add any new information
    - if both tone and light presented at first, then both have strong associations with shock
    - blocking is adaptive, limiting false associations
    - compromised / absent in individuals experiencing hallucinations (as in schizophrenia), hyperassociative tendency, basis for delusional beliefs

6.5 — Learning to Feel With Classical Conditioning
- affect complex emotions
- Little Albert and white lab rat
    - touch rat -> loud noise (US) causing him to cry (UR)
    - 7 pairings: fear of rat
    - after acquisition: rat (CS) produces crying (CR)
    - generalization of fear to other white furry objects (rabbit, fur coat, Santa Claus mask)
    - scientists did nothing to counteract the conditioning

- *counterconditioning* to counteract associations by conditioning new responses (smiling) to stimuli (white rat)
    - Little Albert was removed by mother so this couldn't happen
- negative associations are so powerful because learned rapidly and easily generalized
- PTSD: anxiety for years after the event
    - unimportant stimuli overgeneralized -> leads to fear response
    - world filled with triggers
    - counterconditioning & extinction used in therapy for treatment



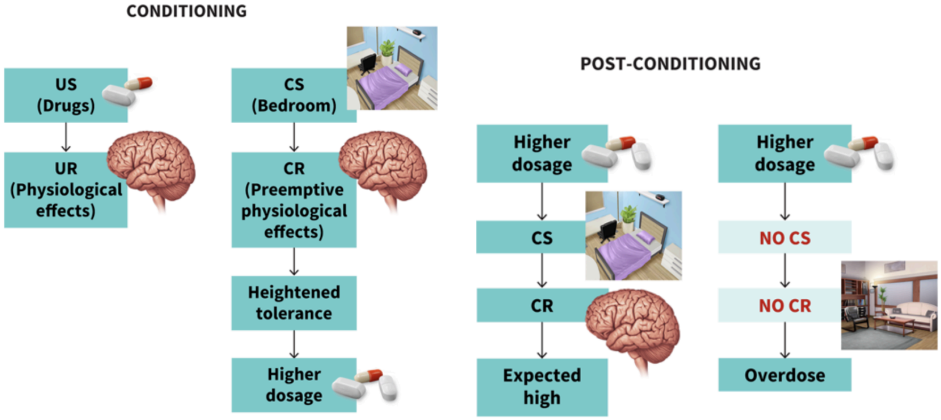
6.6 — Biologically Predisposed for Association

- associations can be learned and handed down from genes
- animals -> internal mechanisms guiding learning
- **preparedness:** species-specific biological predisposition to learn some associations more quickly than other associations
    - e.x. specific phobias
    - spiders, snakes, heights (from ancestors)
- nonhuman animals: monkeys fear snakes
    - *vicarious classical conditioning*
    - observing another organism's experience
- **conditioned taste aversion:** A classically conditioned response where individuals are more likely to associate nausea with food than with other environmental stimuli.
    - future sight & smell of that food -> feel sick & nauseous
    - rat study: sweet water rather than light & clicker became CS for nausea (CR)
- less evolutionarily relevant associations take more trials to be made

6.7 — Classical Conditioning in the Real World

- brain's response to chronic drug use
    - take drugs in same context (paraphernalia & location)
    - environmental stimulus (bedroom) becomes CS
    - CR prepares to protect brain & body, adjusting neurotransmitter levels to counteract drug impact
    - 3 consequences:

        - 1. cravings (feel opposite way drug make feel -> need to take drug to feel normal)
        - 2. drug tolerance (CR anticipates and opposes drug, lessening impact -> progressively less responsive to doses)
        - 3. risk of overdosing
            - in familiar environment (CS), anticipate use of drugs, drug tolerance down
            - in unfamiliar environment, no preemptive adjusting, could overdose (learned drug tolerance associated w/ context absent)



- tool for persuasion
    - sell objects
    - new brand of drink & add attractive people having fun (US) -> brand (CS) -> good feelings (CR)
    - cigarette companies
- conditioned taste aversions to solve practical problems
    - livestock to avoid poisonous plants
    - prevent predators like wolves from attacking livestock
    - counteract bears eating soldiers' meals by putting small dose of nonlethal fungus
    - help cancer patients: scapegoat food

6.8 — Nuts and Bolts of Operant Conditioning

- **operating conditioning:** second form of associative learning in which a learner makes associations between a voluntary behavior and its consequences and makes a behavioral change as a result
- contrast: classical conditioning & INVOLUNTARY reflexes / EMOTIONAL responses
- now, VOLUNTARY actions to control / operate on the environment to produce desired outcomes
- Edward Thorndike: operant conditioning with cats
    - cat-sized "puzzle boxes"
    - door to perform specific action
    - reward: freedom & tasty fish
    - each time cat placed back in the box, less and less random behaviors and string-pulling expected sooner and sooner
    - trial and error to acquire the skill
- **law of effect:** behavior is a function of its consequences
    - actions followed by good outcome = strengthened; vice versa

    - Thorndike: good outcome “satisfiers” and bad “annoyers”
- ABCs:
    - antecedent (stimulus preceding behavior, context): being placed in a box
        - red light: accelerator bad; green light: accelerator good
    - behavior
    - consequence (stimulus after behavior, increases or decreases likelihood of behavior repeating)

6.9 — Your Behavior Has Consequences

- “you reap what you sow”
- consequences -> repeated behavior or not
- satisfiers & annoyers became:
    - **reinforcement:** increased likelihood of behavior being repeated
        - **primary reinforcers:** innately satisfying because meet biological need + effective regardless of prior experience (food, drink, warmth when cold, sex)
        - **secondary reinforcers:** learned, acquiring value through experience because of association with primary reinforcers
            - children: treats, tokens for treats/privileges
            - motivation for work (money)
            - powerful because there’s no limit
                - water useful, diamond higher price
    - **punishment:** decreased likelihood
        - **positive punishment:** unpleasant stimulus presented to decrease likelihood of behavior (scolded by teacher for using cell phone in class)
        - **negative punishment:** decreases behavior by removing stimulus (teacher taking away phone after using it too many times)
- positive (behavior leads to stimulus presented), negative (stimulus removed)
    - **positive reinforcement:** frequency of behavior increases because followed by reinforcing stimulus
    - **negative reinforcement:** increases probability of behavior by avoiding / removing an undesired outcome (remove hunger, thirst, headache, *etc.*)

6.10 — Behavior Modification: Putting Reinforcement to Work for Personal Change

- punishment may not always be appropriate
- children are more strongly engaged when receiving feedback about poor performance (reinforcement)
- adults have less difference in performance in type of feedback received (reinforcement vs punishment)
- praise when children are doing well rather than punishing?
- drug use: InSite (bring drugs to site)
    - not criminalizing but rewards responsible usage
- **Premack principle:** relativity theory of reinforcement
    - activities someone frequently engages in can be harnessed and used to reinforce activities that the person is less inclined to do
    - **preferred behavior**: naturally more reinforcing, engaged in more frequently
    - reinforce less preferred behavior: can’t eat dessert (preferred behavior) until you eat your vegetables (less preferred behavior); can’t go outside (preferred) until homework is finished (less preferred)

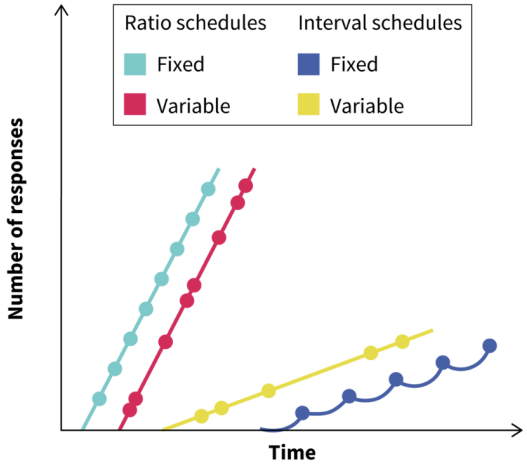
    - not only joy of scrolling reinforcing studying but the fact that it is a frequent behavior
- elapsed time between performance of behaivor & receipt of punishment
    - delayed reinforcement: paycheck, holidays, retirement
    - delayed punishments: grounded, detention, time in prison
- immediate reinforcements can conflict with delayed punishments
    - eat candy -> causes cavities
    - *delay discounting*: underplay importance / extent of delayed consequences
    - guides people towards impulsive decisions & more susceptible to substance use
    - future consequences seem to have less potency

6.11 — Learning New Habits and Breaking Old Ones

- B. F. Skinner: operant chamber (Skinner box) to examine classical and operant conditioning in lab animals
    - pigeon & rat guy
    - small world in which experimenter had control over everything animal experienced
    - allowed for *free operant responses*: allow animals to respond at any time, as many times as needed w/o intervention because no clear end point / goal like to escape
    - measure time to respond & changes in number of responses
    - exhibit reinforced behavior many times to earn many food / water rewards
    - rats to press levers for food (not instinctive, so learned)
- **shaping:** The process by which random behaviors are gradually changed into a desired target behavior.
    - *reinforcement of successive approximations*
    - behavior vaguely similar to target behavior -> complexity -> target behavior achieved
    - reward rat for just taking a step towards lever -> only reward rat for touching lever -> only reward rat when pressing the lever
    - use it to get toddlers to go to the toilet
    - how trainers get animals to do incredible things (click as secondary reinforcer)
- **instinctive drift:** An animal's reversion to evolutionarily derived instinctive behaviors instead of demonstrating newly learned responses. ("old habits die hard")
    - "wild" side of animals: whales, tigers, elephants kill trainers with whom they have worked for years

6.12 — Learning on Schedule

- **continuous reinforcement schedule:** behavior rewarded every time it's performed
    - chocolate every time finish reading a study unit
    - rapid acquisition of behavior
        - but also subject to rapid extinction
        - as soon as no chocolate, no reading
    - example: vending machines
- **partial reinforcement schedules:** behavior rewarded only some of the time
    - very effective motivators
    - slot machine
    - *ratio schedules*: schedules determined by amount of behavior
    - *interval schedules*: determined by time elapsed
    - variable schedule: more randomly & unpredictably (more resistant to extinction than fixed schedules because of unpredictability of reward)

- **fixed-ratio schedules:** A reinforcement schedule in which a specific number of behaviors are required before a reward is given.
- **variable-ratio schedule:** A reinforcement schedule in which an average number of behaviors are required before a reward is given.
    - surprisingly resistant to extinction (slot machine)
- **fixed-interval schedule:** A reinforcement schedule based on a fixed amount of time before a reward is given.
    - drop-off in responding immediately after receiving the reward and goes up once time is almost reached
    - 

- **variable-interval schedule:** A reinforcement schedule based on an amount of time between rewards that varies around a constant average.
    - slow and consistent responses because time to next reinforcer is relatively unpredictable
    - desire to check inbox for email or messages for new text
    - fishing & surfing

6.13 — Beyond Behavior: The Role of the Mind in Learning

- early: learning entirely in terms of observable behavior — *behaviorists*
- *contingent reinforcement*: specific response reinforced because yield desired change
- *noncontingent reinforcement*: reward delivered on fixed-interval no matter what behavior they were doing
- **superstitious conditioning:** whatever behavior birds happened to be doing when first reinforcement arrived was reinforced
    - humans also experience this
    - prediction-machine brains, in trying to extract meaningful patterns out of noise, will invent a pattern when none actually exists
- learning = what we think we have learned
- *cognitive revolution*: uncover underlying mental processes for behavior
    - learning can occur with no outward behavioral evidence
    - **latent learning:** Learning that occurs without either incentive or any clear motivation to learn.
        - naturally form map of surroundings
- Tolman: *cognitive maps*
    - internal mental representation of the world
    - complex spatial layout — the big picture

- operant conditioning has a *learning curve*
- latent learning -> sometime develop mental concept of how things work
- **insight solution:** A form of learning that occurs without trial and error and thus without clear reinforcement.
    - chimps assessing options to get banana and getting it right first try without trial-and-error reinforcement
    - internal concept of the solution

6.14 — Social Learning Theory: Imitation is More Than Just Flattery

- complex human skills
- **observational learning:** observes and imitates a behavior from a model (eliminates painstaking trial-and-error, increasing rate of learning)
- **imitation:** purposeful copying of a behavior
    - infants imitate facial actions
    - first teacher
- **social learning theory:** A theory of how people's cognitions, behaviors, and dispositions are shaped by observing and imitating the actions of others.
    - *attention:* admiration -> attentive to behavior
    - *retention*: storing information in the mind to be used by learner in the future
        - images, verbal descriptions, taking notes, *etc.*
    - *motor reproduction*: reinforce skill as learner physically imitates model's actions
    - *reinforcement*: being praised -> increase likelihood of repetition
- importance of guiding and influencing behavior in children
    - Bobo doll experiment: learning aggression through observation
    - *vicarious reinforcement*: children who saw adult being rewarded for aggression to the doll
    - *vicarious punishment*: saw adult being punished
    - children learned consequences -> behavior

6.15 — The Brain Basis of Observational Learning

- **mirror neurons:** Neurons that are active both when performing an action and when the same actions are observed in others.
    - observe internally and mirror another's behavior without outwardly imitating
    - mentally stimulating observed actions -> perform same actions
- TMS / functional neuroimaging studies support this
- mirror nervous systems allow us to understand what others are doing and perhaps thinking & internal mental states (understanding others' intentions)
    - activated when someone else is in pain (learn from their experiences)

6.16 — Observational Learning as the Basis of Culture

- **cultural transmission:** The transfer of information from one generation to another that is maintained not by genetics, but by teaching and learning.
    - conditioning + observational learning
    - **vertical transmission:** across generations, knowledge and skills (language, food preparation, religious worship, *etc.*)
    - **horizontal transmission:** social learning within the same generation of peers (new technology, fashion, social media rules, *etc.*)
- happens in other species too

- one group to another
- **diffusion chain:** A process in which individuals learn a behavior by observing a model and then serve as models from whom other individuals can learn.
- innovations spread widely and rapidly
    - in humans, language allows more efficient knowledge transmission
    - *cumulative cultural evolution*: continuously building on accumulated skills and knowledge of prior generations
    - *dual inheritance* or *gene-culture coevolutionary theory*: human cultural evolution = dominant force driving species' genetic evolution over last few hundred years
    - brain & cognitive specialization and anatomical changes in guts, teeth, hands, bones